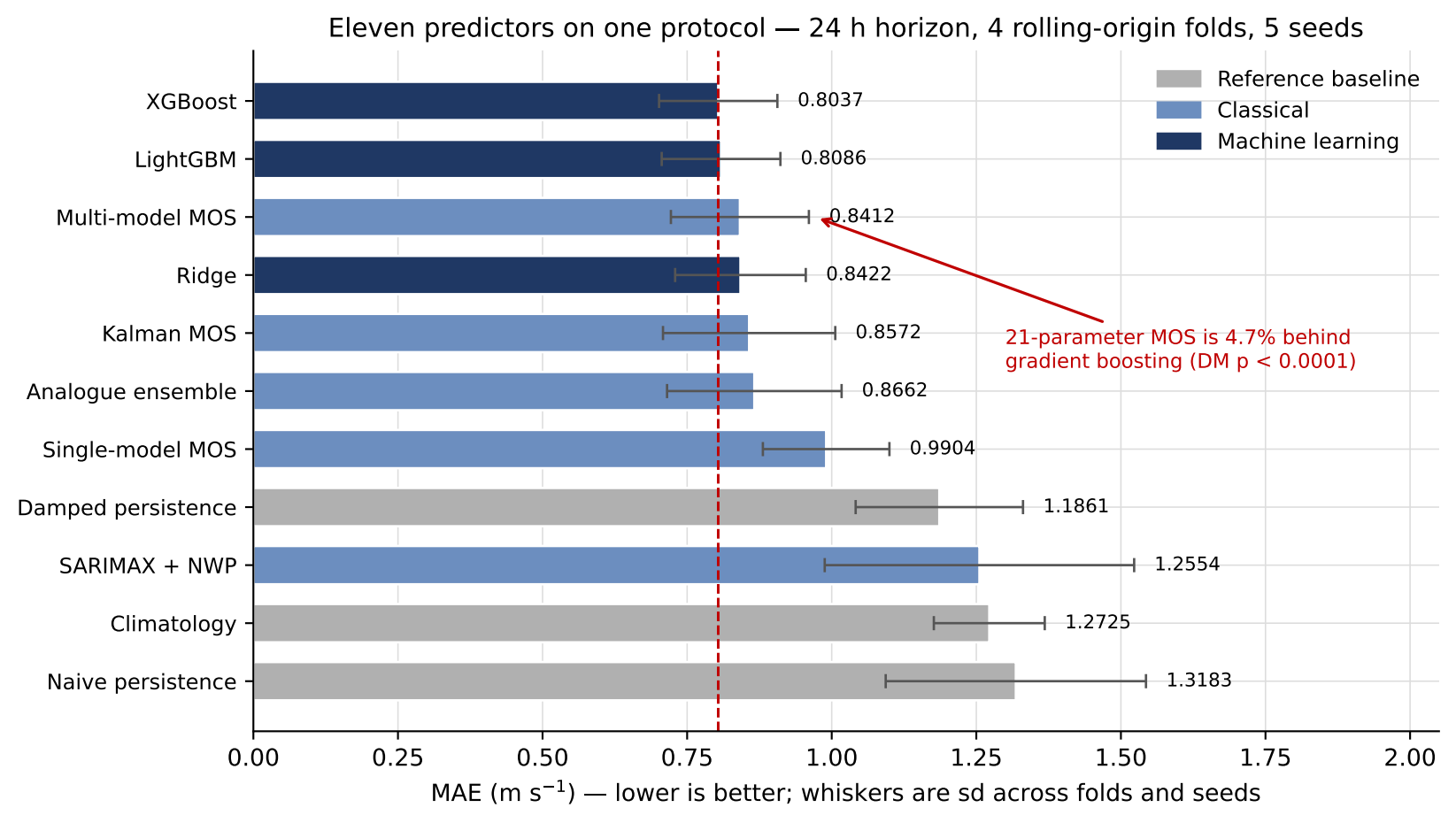
Eleven predictors on one protocol — 24 h horizon, 4 rolling-origin folds, 5 seeds
Reference baseline
Classical
Machine learning
XGBoost
LightGBM
Multi-model MOS
Ridge
Kalman MOS
Analogue ensemble
Single-model MOS
Damped persistence
SARIMAX + NWP
Climatology
Naive persistence
0.8037
0.8086
0.8412
0.8422
0.8572
0.8662
0.9904
1.1861
1.2554
1.2725
1.3183
21-parameter MOS is 4.7% behind
gradient boosting (DM p < 0.0001)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
MAE (m s^{-1}) — lower is better; whiskers are sd across folds and seeds